BALAJI DILIPSINGH RAJPUT

Biotechnology Graduate | Entry-Level QA/QC Trainee | Computational & Wet-Lab Analyst

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CAREER OBJECTIVE & PROFESSIONAL SUMMARY

Goal-oriented and detail-focused Biotechnology graduate with demonstrated academic excellence and foundational competence in cGMP/GDP compliance, wet-lab analytical techniques, and computational pipelines. Recipient of the **Best Research Project Award (2024)** and **Dean's List** distinction. Practical exposure encompasses genomic data workflows (Biotecnika), GLP microbiology & PCR testing (Gujarat Biotech Innovations), and industrial exposure in Oral Solid Dosage manufacturing. Proactive learner seeking an entry-level position as **Quality Assurance (QA) Trainee**, **IPQA Officer**, or **Quality Control (QC) Chemist** in Gujarat's pharmaceutical sector. Available to join immediately.

CORE COMPETENCIES & FOUNDATIONAL KNOWLEDGE

- cGMP, GDP & GLP Principles
- SOP Drafting & Document Lifecycle
- In-Process Quality Checks (IPQA)
- Good Documentation Practices (ALCOA+)
- 21 CFR Part 11 & Schedule M Awareness
- Label Reconciliation & Inspection
- Deviation & CAPA Basics
- BMR / BPR Review Protocols
- HPLC & Spectrophotometry Concepts
- Microbial Testing & Aseptic Culture
- Quality Management System (QMS)
- Root Cause Analysis (5-Why, Fishbone)

EDUCATION

Diploma in Biotechnology **2021 – 2025**

Parul Institute of Technology & Engineering, Gujarat Technological University (GTU), Vadodara

Honors / Performance: CGPA: 6.7 / 10 | Dean's List for Academic Excellence (2024–2025)

Key Subjects: Molecular Biology, Microbiology, Bioprocess Technology, Biochemistry, Immunology, Fermentation Technology, QC & QA.

INDUSTRIAL TRAINING & INTERNSHIP EXPERIENCE

Industrial QA Trainee / Apprentice — Elysium Pharmaceuticals Ltd., Dabhasa, Vadodara **Mar 2024 – Mar 2026**

- Received hands-on industrial training in Oral Solid Dosage (tablets) manufacturing under strict cGMP and Schedule M standards.
- Assisted IPQA officers in line clearance, physical parameter checks (hardness, friability, disintegration time), and BMR/BPR log verification.
- Observed and supported SOP authoring, deviation tracking, and preparation of documentation packages for CDSCO/WHO-GMP audit readiness.

Bioinformatics Intern — Biotecnika (Remote) **2024 (3 Months)**

- Engineered automated Python / Biopython workflows for sequence alignment, screening over 500+ DNA and protein sequences.
- Utilized BLAST, ClustalW, and GenBank annotations to streamline curation workflows, cutting manual processing time by 40%.

Laboratory Research Intern — Gujarat Biotech Innovations **2023 (2 Months)**

- Carried out routine laboratory assays: PCR, agarose gel electrophoresis, and microbial culture handling for 80+ samples/week.
- Authored 3 standard operating procedures (SOPs) compliant with GLP standards; elevated laboratory sample workflow efficiency by 20%.

KEY ACADEMIC & RESEARCH PROJECTS

Bioethanol Production from Agricultural Waste (2024) **[Fermentation, DOE/RSM, HPLC, Microbial Kinetics, ANOVA]**

Optimized Simultaneous Saccharification & Fermentation (SSF) with *S. cerevisiae* via Response Surface Methodology (RSM/DOE); achieved a **15% yield enhancement** validated by HPLC and ANOVA ($p < 0.05$). **Awarded Best Research Project – Parul Institute 2024**.

Comparative Genomic Analysis of AMR Genes (2024) **[Python, Biopython, BLAST, GenBank, ClustalW, Bioinformatics]**

Investigated 120 bacterial genomes from NCBI GenBank using an automated Python/Biopython workflow; identified **8 novel Antimicrobial Resistance (AMR) gene variants** across 500+ sequence entries.

Protein Structure Prediction & Molecular Docking (2023) **[SWISS-MODEL, AutoDock Vina, PyMOL, PDB, In Silico Drug Design]**

Constructed 3D homology models for 5 therapeutic target proteins using SWISS-MODEL; performed virtual screening and molecular docking with 50 ligands via AutoDock Vina, identifying 3 lead inhibitors (binding affinity ≤ -8.0 kcal/mol).

Biogenic Synthesis of Silver Nanoparticles (AgNPs) (2023) **[Green Synthesis, UV-Vis Spectrophotometry, Microbial Assays, Aseptic Techniques]**

Synthesized antimicrobial AgNPs via green biogenic route using *Pseudomonas spp.*; confirmed nanoparticle formation and broad-spectrum bactericidal efficacy via UV-Vis Spectrophotometry and zone-of-inhibition assays.

TECHNICAL & LABORATORY SKILLS

Quality & Regulatory Awareness: cGMP, GDP, GLP, Schedule M, SOP Drafting, BMR / BPR Verification, Deviation & CAPA Basics, Change Control Awareness, Data Integrity (ALCOA+), 21 CFR Part 11, CDSCO & WHO-GMP Guidelines.

Laboratory & Analytical: HPLC (Fundamentals), PCR / RT-PCR, UV-Vis Spectrophotometry, Agarose Gel Electrophoresis, Microbial & Aseptic Culture, Sterility Testing, Disintegration & Friability Testing, Autoclave, LAF.

Bioinformatics & Computation: Python (Biopython, Pandas, NumPy), R Basics, BLAST, ClustalW, NCBI GenBank, UniProt, AutoDock Vina, SWISS-MODEL, PyMOL.

Software & Digital Tools: MS Office Suite (Advanced Excel, Word, PowerPoint), SPSS, MATLAB (Basic), Git & GitHub, TrackWise (Basics), Google Workspace.

CERTIFICATIONS & SPECIALIZED TRAINING

- Bioinformatics Specialization – Coursera / UC San Diego (2024)
- Python for Genomic Data Science – Coursera / Johns Hopkins (2024)
- Industrial Biotechnology & GMP – NPTEL / IIT Madras (2023)
- Advanced Techniques in Biotechnology – Parul University (2025)
- Molecular Biology Techniques – Udemy (2023)
- Digital Marketing & Web Foundations – Google / ITI (2021–2024)

KEY HONORS & ACHIEVEMENTS

- **Best Research Project Award (2024)** – Parul Institute of Technology & Engineering for Bioethanol SSF optimization.
- **Dean's List for Academic Excellence (2024–2025)** – Recognized for top percentile performance in Biotechnology coursework.
- **Automated Genomic Pipeline** – Developed Python workflow reducing manual sequence curation time by 40% (AMR study).

PERSONAL DOSSIER & DECLARATION

Languages Known: English (Fluent), Hindi (Fluent), Gujarati (Native)
Details: Date of Birth: 20 June 2001 | Marital Status: Single | Nationality: Indian | Availability: Immediate Joiner

I hereby solemnly declare that all statements and particulars provided in this resume are true, complete, and authentic to the best of my knowledge.

Place: Vadodara, Gujarat
Date: _____

(Balaji Dilipsingh Rajput)
Candidate Signature